

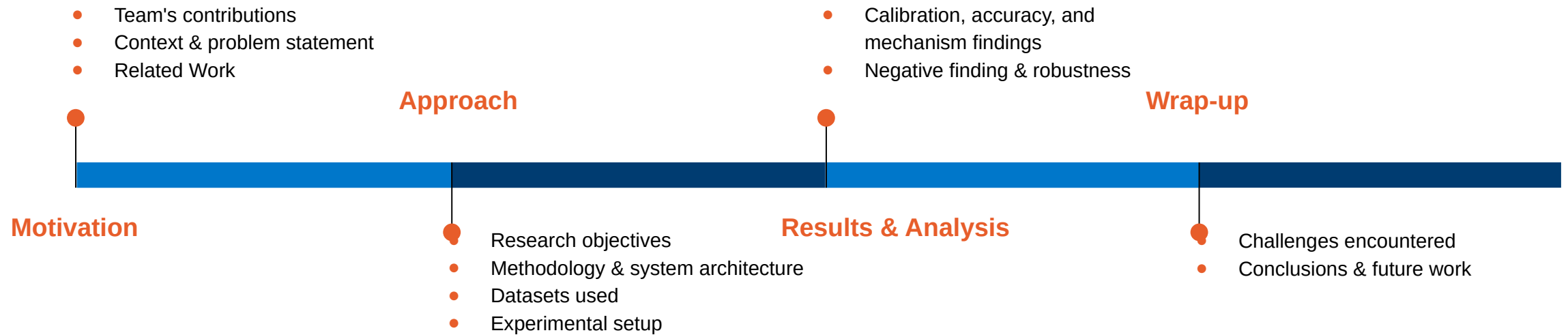
IoT Thunders Team

Mustafa Eren Soyhan	101045056
Mohamed Ahmed	101031735
Sudha Rajendran	101015699
Thuy Trang Cao	101015203
Vaidehi Patel	101013014

Supervised by **Professor Khalid Elgazzar**
Proposal by **Mohamed Hafez**

Uncertainty-Weighted Test-Time Augmentation for Robust Medical Image Classification

Agenda



Team Members' Contributions

S1 Mustafa Eren Soyhan

Repository admin & pipeline owner with core TTA + fusion code, implemented and ran all of EfficientNet-B0 and DeiT-Tiny backbones, Top-K strategies, unified runner & inference benchmarking, report and conference submission review.

S2 Mohamed Ahmed

DermaMNIST Owner. Publication tables, figures, and bootstrap confidence intervals; Figure 1 “Augmentation Confidence Strips” composite assembler, bug fixing the pipeline, report and conference submission review

S3 Thuy Trang Cao

BreastMNIST owner. Executed 4-phase pipeline: baselines, standard/weighted TTA (8 strategies), ablations, and 3-seed stability analysis. Worked on results, discussion, conclusion & selection guide

S4 Vaidehi Patel

OrganAMNIST owner. Baseline + weighted-TTA experiments; results tables & reliability diagrams, report methodology, dataset explanation, experimental setup

S5 Sudha Rajendran

PneumoniaMNIST owner. Executed full pipeline: baselines, 8 uncertainty-weighted TTA strategies, and multi-seed ablation studies. Worked on Introduction, related work, and references, contextualizing our approach within existing medical image calibration literature.

Mohamed Hafez SUPERVISOR

Project Proposal & Mentorship: Provided the foundational proposal, expert guidance, and invaluable feedback that shaped and elevated this research.

Context & Problem Statement

Context. Medical-image classifiers are usually deployed with a single prediction per image and those predictions are often **overconfident** and **poorly calibrated**, so the model's stated confidence does not match how often it is actually right.

- **Why it matters.** In clinical decision support this is **risky**: a clinician may over-trust an **overconfident AI**.
- **Our question.** When does TTA **help** or **hurt** calibration and can weighting each view by the model's uncertainty give a more reliable alternative?

Test-Time Augmentation (TTA) runs N augmented versions of the same image through the model, then fuses their predictions without retraining.

But a 2026 study (*"I Can't Believe TTA Is Not Better"*) showed equal-weight averaging often **hurts** as bad views (heavy brightness, wrong flips) mislead the vote.

88.1% → 56.5%

PathMNIST accuracy collapse with 50 equal-weight TTA views
(Medeiros (2026) - prior work)

a 31.6 percentage-point (pp) degradation

Related Work

Yang et al. (2023)

"MedMNIST v2: A Large-Scale Lightweight Benchmark"

Our baseline evaluation fully replicates the published ResNet-18 benchmark limits across all modalities.

Medeiros D. N (2026)

"I Can't Believe TTA Is Not Better"

Showed that equal-weight averaging over medical-image augmentations can **hurt** when distorted views, such as heavy brightness or wrong flips, mislead the vote.

Gao et al. (2022)

"A Comparative Study of Confidence Calibration"

Shows that calibration behavior can differ between natural-image and medical-image settings.
Simpler CV calibration strategies generalize much better to the unique data distributions of medical imaging.

Research gap

- Standard equal-weight TTA can harm medical classifiers.
- Uncertainty-weighted TTA has never been studied across multiple medical modalities.
- **We fill that gap on 6 datasets × 3 backbones.**

Our idea

- Weight each view by how confident the model is about it. Confident views count more; uncertain views count less.
- We test whether this recovers the loss and more importantly improves calibration (ECE).

Research Objectives

1 Replicate the failure

Show standard equal-weight TTA degrades accuracy / calibration across diverse medical modalities.

2 Propose weighted fusion

Evaluate uncertainty-weighted strategies: entropy, max-prob, predictive variance, MC-Dropout.

3 Measure calibration, not just accuracy

Report Expected Calibration Error (ECE) and NLLas clinically relevant confidence-quality metrics

4 Find modality patterns

Identify which modalities benefit and which augmentations are safe per modality.

5 Agnostic toolkit

Model-agnostic pipeline to be applied to any trained PyTorch classifier in a few lines, with no retraining.

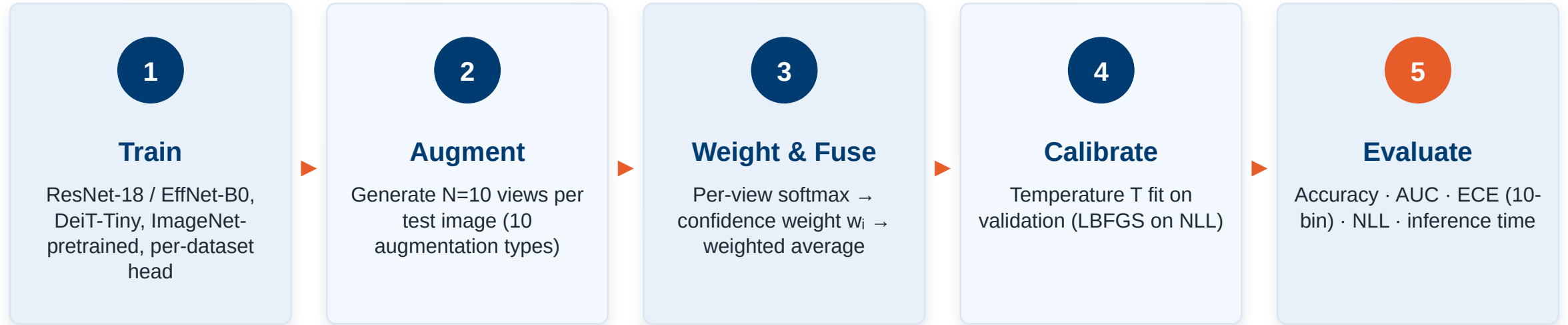
Approach

Presented by: Mohamed Ahmed



Methodology & System Pipeline

One trained model → N augmented views → confidence-weighted fusion → calibrated prediction



The core idea, formally

Standard TTA: $\bar{p} = \frac{1}{N} \sum_i p_i$ **Ours:** $\bar{p} = \frac{\sum_i w_i p_i}{\sum_i w_i}$ where w_i reflects the model's confidence on view i .

One unified runner orchestrates train → TTA → calibrate → benchmark across three backbones, three seeds, and all datasets from one command.

Fusion Strategies Compared

Eight main fusion strategies are shown below; Top-K entropy filtering is reported separately as an ablation.

Baseline TTA

Every view weighted equally

$$w_i = \frac{1}{N}$$

Variance-inv

Literal proposal formula → negative finding

$$w_i = \frac{1}{var + \epsilon}$$

Max-Prob

Confident views dominate

$$w_i = \max(p_i)$$

MC-Dropout

Epistemic proxy, no augmentation

T stochastic passes

Entropy

Low-entropy views weigh more

$$w_i = e^{-H(p_i)}$$

TS-only

Temperature scaling, single pass which is nearly free

softmax(z / T)

Variance

Sharp views up (confidence-aligned)

$$w_i = \text{var}(p_i)$$

TS + Entropy

Safest NLL default; often near-best ECE

entropy on z / T

Datasets Used

Dataset	Modality	Task	Classes	Train size
PathMNIST	Colon pathology (histology)	Multi-class	9	~90,000
DermaMNIST	Dermatoscope (skin)	Multi-class	7	~7,000
PneumoniaMNIST	Chest X-ray	Binary	2	~4,700
BreastMNIST	Breast ultrasound	Binary	2	546
BloodMNIST	Blood-cell microscopy	Multi-class	8	~12,000
OrganAMNIST	Abdominal CT	Multi-class	11	~35,000

All from MedMNIST v2 · 64×64 · pip-installable & pre-split · chosen to span 5 modalities and 6 datasets, binary + multi-class, and a wide size range (546 → ~90k) for a diverse benchmark.

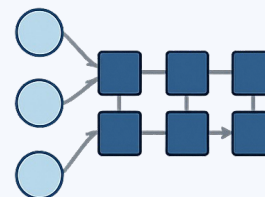
Technology Stack & Experimental Setup

Tech stacks

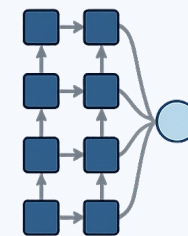
LIBRARIES



MODELS · ImageNet-pretrained



CNNS: ResNet-18 - EfficientNet-B0



ViT: DeiT-Tiny

Training protocol — identical for every dataset & backbone

- Temperature T fit on val (LBFGS); 3 seeds (0 / 42 / 123)
- Adam, lr = 1e-4, 30 epochs, batch = 64, 64×64 input
- Grayscale replicated to 3-channel, ImageNet normalization
- Significance: McNemar (per-dataset) + Wilcoxon + bootstrap CI

Results & Analysis

Presented by: Sudha Rajendran and Vaidehi Patel



Key Evaluation Metrics

Expected Calibration Error (ECE)

Our primary metric, calculated over 10 bins. Measures the gap between predicted confidence and actual accuracy; lower is better. We report it before and after to show how uncertainty-weighted TTA improves calibration.

AUC - ROC

Measures ranking / discrimination; how well the model separates classes; higher is better. Reported alongside accuracy.

Negative Log-Likelihood (NLL)

Measures probability quality and penalizes confident wrong predictions; lower is better.

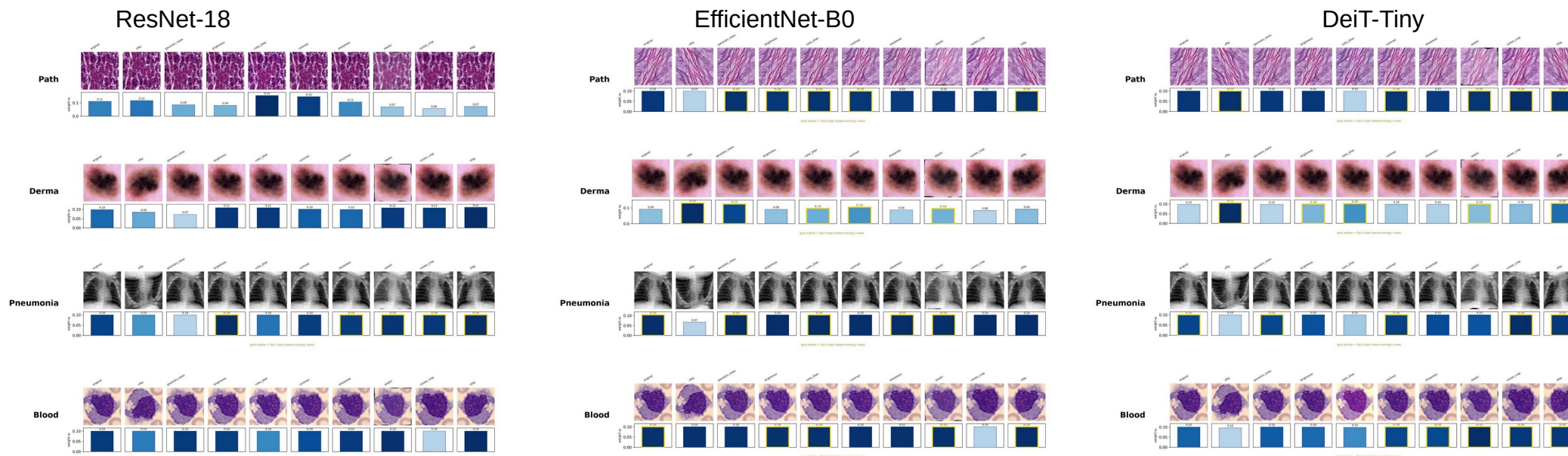
Top-1 Accuracy

The percentage of images where the model's single most confident prediction is correct; higher is better.

Inference Time

Tracks computational efficiency in ms/image
Recorded post-GPU warm-up using highly precise timing
(`time.perf_counter()`)

Augmentation Confidence Strips



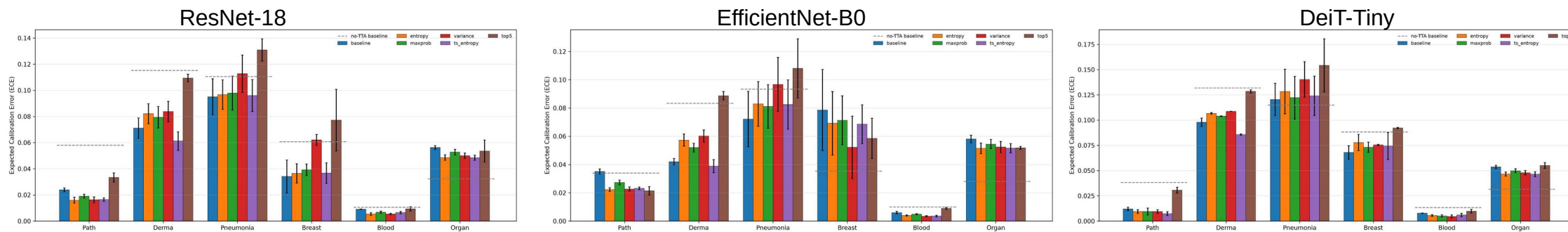
Visual Guide

- Each strip shows 10 augmented copies of one test image.
- Bar below each that shows how much the entropy ensemble trusts that view (taller / darker = more trusted).

Takeaways

- The model down-weights distorted views on its own as no labels needed.
- Trust is modality-specific: the same augmentation can receive different confidence weights across modalities.
- This is the paper's Figure 1 - the VMV visual contribution. Strips exist for all 6 datasets, 3 backbones.

Calibration Is Where We Win



Best uncertainty-aware strategy vs equal-weight TTA (3-seed mean) representative subset of the 9 strategies; dashed = no-TTA baseline

Takeaways

- TS + Entropy is the safest NLL default and is often near-best on ECE; TS-only is strongest on OrganAMNIST.
- Naïve equal-weight TTA leaves real miscalibration on the table and can make a calibrated model worse.
- The clearest gains appear on selected multi-class modalities.
- Top-K hard filtering is not a reliable substitute; soft weighting is the better default.
- The same qualitative pattern holds on EfficientNet-B0 and DeiT-Tiny.

-19%

ECE · DermaMNIST

-41%

ECE · BloodMNIST

-33%

ECE · PathMNIST

Accuracy Is a Deliberate Null

Across 6 datasets, weighted TTA does not move accuracy and our statistics say so. The contribution is calibration, not a few extra correct pixels.

+0.04 pp

mean accuracy change
(Wilcoxon, 6 datasets)

p = 0.165

Wilcoxon signed-rank
not significant

$\leq 2 / 6$

datasets with a significant
McNemar accuracy gain

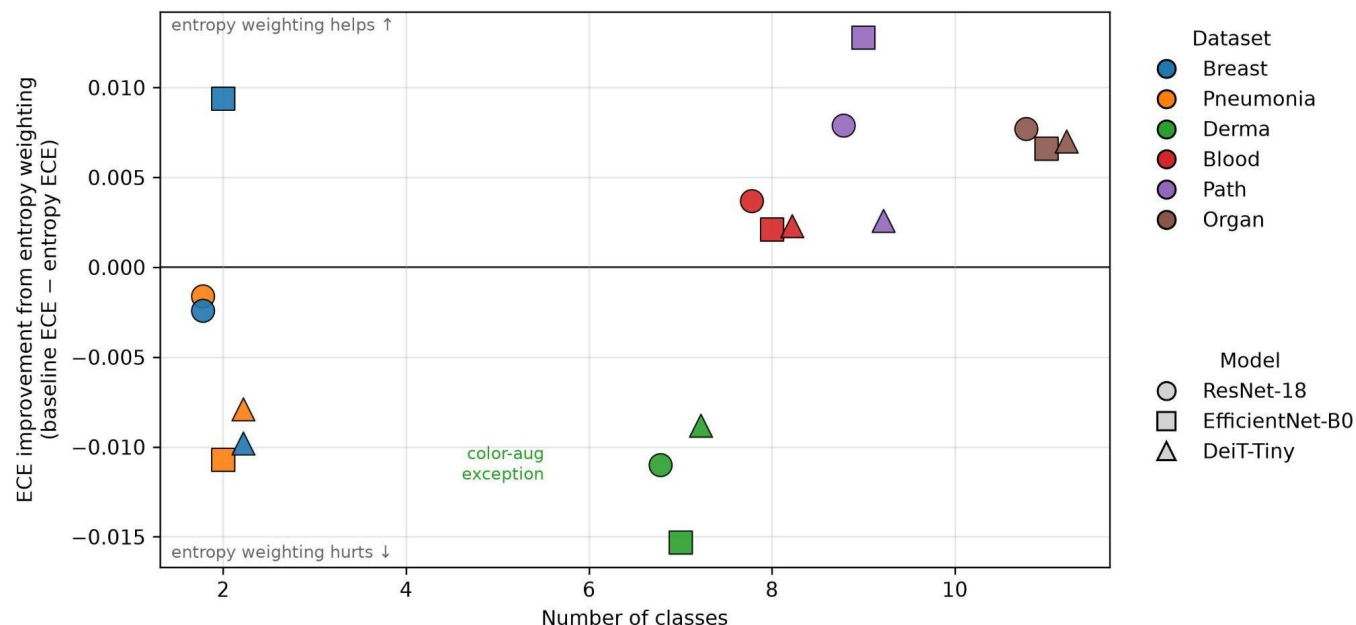
Why a null is the right answer

Prior work showed equal-weight TTA can **destroy** accuracy (−31.6 pp). Our weighting **preserves** accuracy while fixing calibration exactly the safe, deployable behaviour a clinician wants.

...and it's nearly free

TS-only calibrates with a **single forward pass** (≈0.4–2 ms/image vs ~6–60 ms for 10-view TTA). Weighted TTA reuses per-view logits, so weighting adds **no extra cost** over equal-weight.

Why It Works: The Mechanism



Entropy weighting tends to help more on higher-class-count tasks. Binary tasks (Breast, Pneumonia) gain little; Derma is the colour-aug exception (below the line).

The pattern

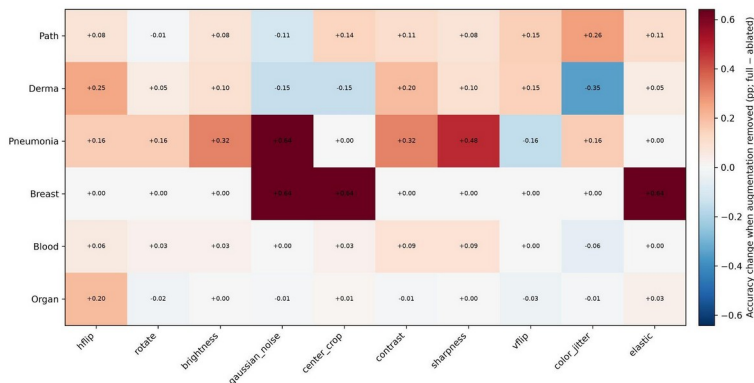
More classes tend to create more view disagreement → more room for confidence weighting to suppress bad views.

Binary tasks (Breast, Pneumonia) have little to gain as they sit near zero.

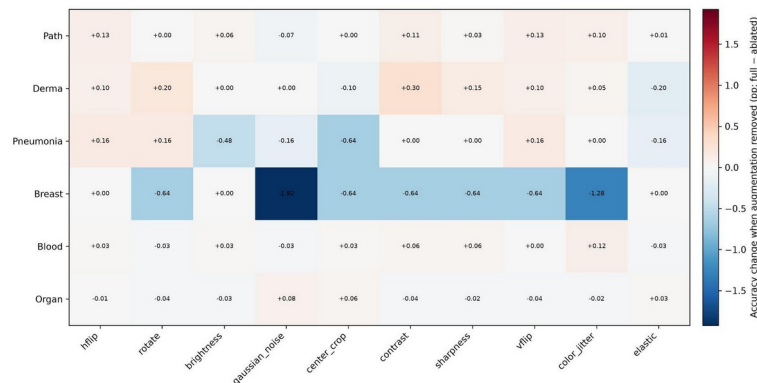
The exception: DermaMNIST

Colour-jitter views look confident but are wrong, so pure entropy weighting **back-fires** on dermoscopy. **Temperature scaling fixes it** - TS + Entropy rescues DermaMNIST relative to pure entropy and gives the safest probability quality.

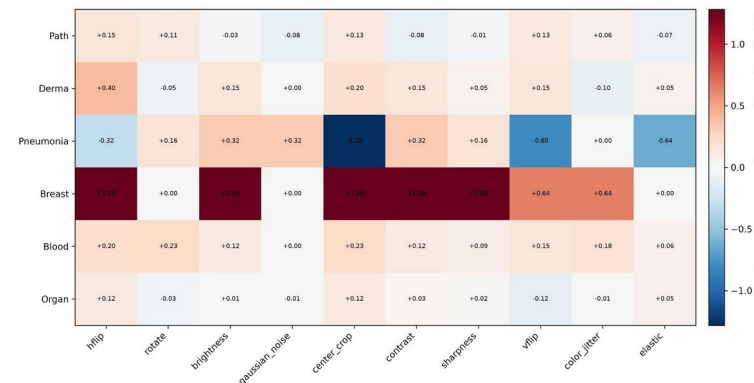
Negative Finding & Safety Guide



ResNet-18



EfficientNet-B0



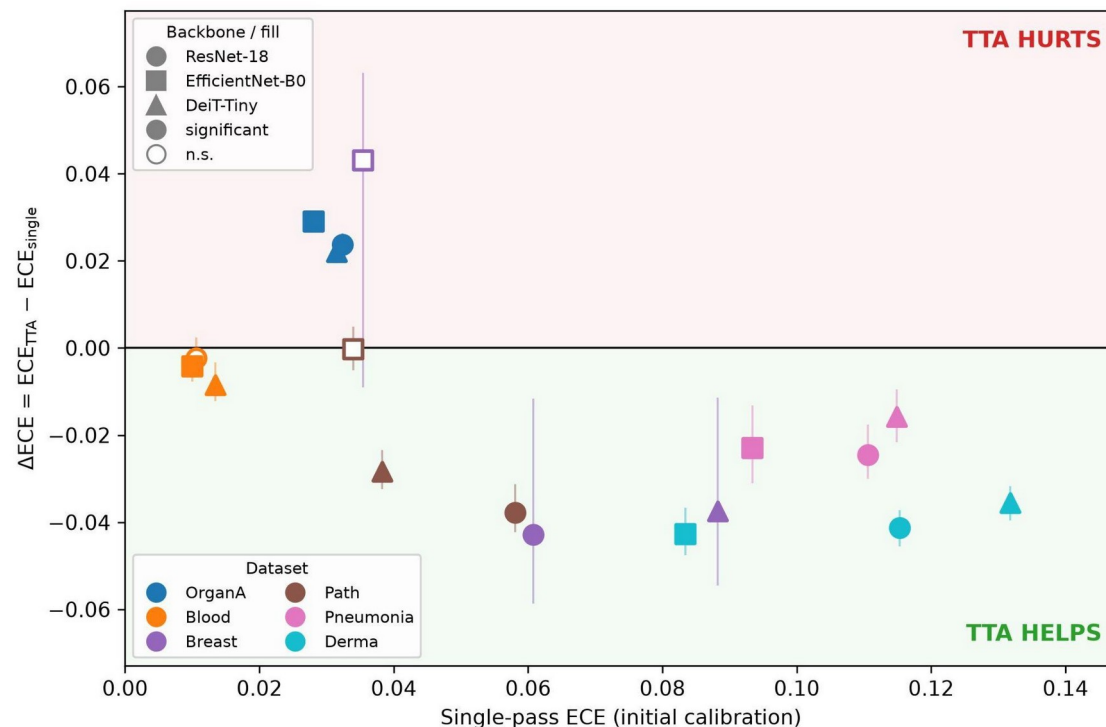
DeiT-Tiny

Leave-one-out augmentation importance (ResNet-18 / EffNet-B0 / DeiT-Tiny): red = removing it hurts accuracy (important aug), blue = removing it helps (harmful aug). Tiny Breast/Pneumonia cells (± 0.64 = one image) are small-sample noise.

What we learned

The literal proposal formula $w = 1/(\text{var} + \epsilon)$ is backwards. Because a confident, peaked prediction has high variance, $1/(\text{var} + \epsilon)$ actually up-weights the uncertain views and it's unstable, dropping BreastMNIST accuracy from 0.84 to 0.77 in the EfficientNet-B0 result. We therefore adopt the confidence-aligned $w = \text{var}$ as our headline strategy and report the literal formula only as a documented negative result.

Why It Works & Where – Mechanism + Augmentation Importance



Augmentation	X-ray (Chest)	CT (Abdominal)	Dermoscopy (Skin)	Microscopy (Histology)	Microscopy (Blood)	Ultrasound (Breast)
Horizontal flip	✗ Avoid	✗ Avoid	△ Caution	✓ Safe	✓ Safe	△ Caution
Vertical flip	✗ Avoid	✗ Avoid	✓ Safe	✓ Safe	✓ Safe	✗ Avoid
$\pm 10^\circ$ rotation	✓ Safe	✓ Safe	✓ Safe	✓ Safe	✓ Safe	✓ Safe
Brightness $\pm 15\%$	✓ Safe	✓ Safe	△ Caution	△ Caution	✓ Safe	✓ Safe
Gaussian noise	✓ Safe	✓ Safe	✓ Safe	✓ Safe	✓ Safe	✓ Safe
Centre crop 90%	✓ Safe	✓ Safe	✓ Safe	✓ Safe	✓ Safe	△ Caution
Contrast $\pm 10\%$	✓ Safe	✓ Safe	△ Caution	✓ Safe	✓ Safe	✓ Safe
Colour jitter	✗ Avoid	✗ Avoid	△ Caution	✓ Safe	✓ Safe	✗ Avoid
Elastic deform	△ Caution	△ Caution	△ Caution	✓ Safe	✓ Safe	△ Caution
Sharpness $\pm 20\%$	✓ Safe	✓ Safe	✓ Safe	✓ Safe	✓ Safe	✓ Safe

There are ten augmentation types: horizontal and vertical flips, rotation ($\pm 10^\circ$), brightness ($\pm 15\%$), additive Gaussian noise ($\sigma=0.01$), centre crop (90% then resize), contrast ($\pm 10\%$), colour jitter (brightness/contrast/saturation $\pm 10\%$, hue ± 0.05), elastic deformation ($\alpha=20$, $\sigma=4$), and sharpness ($\pm 20\%$); stochastic parameters are resampled per view. The clinical appropriateness of these augmentations varies by modality.

Mechanism Takeaway: TTA helps where the model starts poorly calibrated, and hurts only where it's already well-calibrated. x = single-pass ECE, y = ΔECE (TTA – single); filled = significant (95% CI excludes 0); shape = backbone. OrganAMNIST is the only consistent "TTA hurts" case.

Inference Time Per Model

Inference cost

it is given as ms/image, computed on a RTX 4080 Mobile. The single forward pass is pure GPU compute with a fixed batch of 64; end-to-end TTA is wall-clock on the PathMNIST test set, dominated by the backbone-independent augmentation pipeline is about 1 ms per added view, so it scales with N, is similar across backbones, and reaches about 45–55 ms at N=50. The soft strategies and Top-K filters reuse one set of per-view forward passes, so all share the TTA cost at a given N; only MC-Dropout, with T =20 sequential stochastic passes, separates the backbones.

Method	ResNet-18	EffNet-B0	DeiT-Tiny
Single forward pass (compute), ms	0.05	0.13	0.06
TTA end-to-end, N = 5 (ms)	5.32	5.14	6.39
TTA end-to-end, N = 10 (ms)	10.79	9.32	12.97
MC-Dropout, T = 20 (ms)	9.88	11.07	83.30
Marginal cost / added view (ms)	1.09	0.84	1.32
Throughput at N = 10 (img/s)	93	107	77

Wrap-up

Presented by: Thuy Trang Cao



Robustness — Three Backbones, Three Seeds

1 Two Additional Backbones

EfficientNet-B0 & DeiT-Tiny

Reproduces the same overall pattern:
TS+Entropy is the safest NLL
default; inverse-variance is unstable.
The qualitative finding is consistent
across architectures.

2 Seed Stability

3 seeds (0/42/123)

Tight error bars: e.g. PathMNIST
accuracy std ≈ 0.008 , ECE std ≈ 0.002 .
Trends survive seed noise.

3 Top-K Ablation

top3 / top5 / top7

Hard entropy filtering can help some
accuracy cases, but it worsens
probability quality overall; soft
weighting is the better default.

Bottom line: the same qualitative calibration pattern holds across ResNet-18, DeiT-Tiny, and EfficientNet-B0, three seeds, and a 1 $\rightarrow$ 50 view sweep, so the recommendation (use TS + Entropy; avoid 1/var) is robust, not a single-run artefact.

Challenges Encountered

1 Contradiction in the proposal

The variance formula $1/(\text{var}+\epsilon)$ contradicted its own “confident views dominate” intuition. Resolved by supervisor decision: headline $w = \text{var}$; ship $1/(\text{var}+\epsilon)$ as a labelled negative finding.

2 Reproducibility across 5 machines

Five students, five environments. Protected main + pull-request workflow + CODEOWNERS, and one runner computing ECE identically (10 bins), so every row is comparable.

3 Windows DataLoader stalls

Spawned workers reboot CUDA DLLs and hang for minutes. Fixed by defaulting to single-process loading (`num-workers = 0`) on Windows.

4 Small, noisy datasets

BreastMNIST has only 546 training images where single runs are unstable. Required 3-seed evaluation with reported mean $\pm$ std to trust the numbers.

5 EfficientNet MC-Dropout

Untrained EffNet-B0 outputs zero penultimate features in eval mode
→ MC-Dropout shows no spread. Pinned by a test so it isn't mistaken for a bug.

6 Framing accuracy as a null

Our gain is calibration, not accuracy. Needed careful significance testing (McNemar, Wilcoxon, bootstrap CIs) to state the null honestly and defensibly.

Conclusions & Future Work

1 Calibration, not Accuracy

Uncertainty-weighted TTA leaves accuracy unchanged (a deliberate, tested null) while improving ECE most on selected multiclass modalities, with TS + Entropy as the safest NLL default.

2 A clear recommendation

TS + Entropy is the most consistent default, especially by NLL across multiple datasets and backbones. Soft weighting is more reliable than hard Top-K filtering, and the inverse variance formula should be avoided

3 Honest, reproducible science

Results are fully validated across five environments using locked three seeds evaluation, automated unit tests, and identical evaluation metrics (e.g., strictly 10-bin ECE), supporting reproducible and robust findings.

Why it matters for diagnostics: Trustworthy confidence enables safe triage so confident cases can be handled automatically, uncertain ones flagged for a clinician at near-zero added cost with TS-only.

Future work

- Larger backbones & real hospital data (beyond 64×64)
- Edge / streaming deployment TS-only is the cheap path
- Train the model to automatically score the quality of each augmented view
- Add mathematical guarantees to our confidence scores for safer clinical use

References

- <https://medmnist.com/>
- [AB18] AYHAN M. S., BERENS P.: Test-time data augmentation for estimation of heteroscedastic aleatoric uncertainty in deep neural networks. In Proceedings of the International Conference on Medical Imaging with Deep Learning (MIDL) (2018). URL: <https://openreview.net/forum?id=rJZz-knjz>.
- [ALMV20] ASHUKHA A., LYZHOV A., MOLCHANOV D., VETROV D.: Pitfalls of in-domain uncertainty estimation and ensembling in deep learning. In Proceedings of the International Conference on Learning Representations (ICLR) (2020). doi:[10.48550/arXiv.2002.06470](https://doi.org/10.48550/arXiv.2002.06470).
- [APH*21] ABDAR M., POURPANAH F., HUSSAIN S., REZAZADEGAN M., LIU L., GHAVAMZADEH M., FROOM P., ALIZADEHSANI R., KHASHEI M., YIN S., ET AL.: A review on uncertainty quantification in deep learning: Techniques, applications and challenges. Information Fusion 76 (2021), 243–297. doi:[10.1016/j.inffus.2021.05.008](https://doi.org/10.1016/j.inffus.2021.05.008).
- [BLBA22] BATESON M., LOMBAERT H., BEN AYED I.: Test-time adaptation with shape moments for image segmentation. In Proceedings of the International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI) (2022), pp. 736–745. doi:[10.1007/978-3-031-16440-8_70](https://doi.org/10.1007/978-3-031-16440-8_70).
- [GLT*22] GAO R., LI T. Z., TANG Y., XU Z., KAMMER M., ANTIC S. L., SANDLER K., MALDONADO F., LASKO T. A., LANDMAN B. A.: A comparative study of confidence calibration in deep learning: From computer vision to medical imaging. arXiv preprint arXiv:2206.08833 (2022). doi:[10.48550/arXiv.2206.08833](https://doi.org/10.48550/arXiv.2206.08833).
- [KG22] KARIMI D., GHOLIPOUR A.: Improving calibration and out-of-distribution detection in deep models for medical image segmentation. IEEE Transactions on Artificial Intelligence 4, 2 (2022), 383–397. doi:[10.1109/TAI.2022.3159510](https://doi.org/10.1109/TAI.2022.3159510).
- [KHM*22] KURZ A., HAUSER K., MEHRTENS H. A., KRIEGHOFF HENNING E., HEKLER A., KATHER J. N., FRÖHLING S., VON KALLE C., BRINKER T. J.: Uncertainty estimation in medical image classification: Systematic review. JMIR Medical Informatics 10, 8 (2022), e36427. doi:[10.2196/36427](https://doi.org/10.2196/36427).
- [ZDKZ22] ZHANG L., DENG Z., KAWAGUCHI K., ZOU J.: When and how mixup improves calibration. In Proceedings of the 39th International Conference on Machine Learning (ICML) (2022), pp. 26135–26160. doi:[10.48550/arXiv.2102.06289](https://doi.org/10.48550/arXiv.2102.06289).
- [KPNK*19] KULL M., PERELLO NIETO M., KÄNGSEPP M., SILVA FILHO T., SONG H., FLACH P.: Beyond temperature scaling: Obtaining well-calibrated multi-class probabilities with dirichlet calibration. In Advances in Neural Information Processing Systems (NeurIPS) (2019), vol. 32. doi:[10.48550/arXiv.1910.12656](https://doi.org/10.48550/arXiv.1910.12656).
- [Med26] MEDEIROS D. N.: I can't believe it is not better: When test-time augmentation hurts medical image classification. arXiv preprint arXiv:2604.09697 (2026). doi:[10.48550/arXiv.2604.09697](https://doi.org/10.48550/arXiv.2604.09697).
- [MMKF*20] MOSHKOV N., MATHE B., KERTESZ-FARKAS A., HOLLANDI R., HORVATH P.: Test-time augmentation for deep learning-based cell segmentation on microscopy images. Scientific Reports 10,1 (2020), 5068. doi:[10.1038/s41598-020-61808-3](https://doi.org/10.1038/s41598-020-61808-3).
- [OFR*19] OVADIA Y., FERTIG E., REN J., NADO Z., SCULLEY D., NOWOZIN S., DILLON J., LAKSHMINARAYANAN B., SNOEK J.: Can you trust your model's uncertainty? evaluating predictive uncertainty under dataset shift. In Advances in Neural Information Processing Systems (NeurIPS) (2019), vol. 32. doi:[10.48550/arXiv.1906.02530](https://doi.org/10.48550/arXiv.1906.02530).
- [RGA22] RAJARAMAN S., GANESAN P., ANTANI S.: Deep learning model calibration for improving performance in class-imbalanced medical image classification tasks. PLOS ONE 17, 1 (2022), e0262838. doi:[10.1371/journal.pone.0262838](https://doi.org/10.1371/journal.pone.0262838).
- [SBBG21] SHANMUGAM D., BLALOCK D. W., BALAKRISHNAN G., GUTTAG J.: Better aggregation in test-time augmentation. In Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) (2021), pp. 1194–1203. doi:[10.1109/ICCV48922.2021.00125](https://doi.org/10.1109/ICCV48922.2021.00125).
- [SNKB23] SAMBYAL A. S., NIYAZ U., KRISHNAN N. C., BATHULA D. R.: Understanding calibration of deep neural networks for medical image classification. Computer Methods and Programs in Biomedicine 242 (2023), 107816. doi:[10.1016/j.cmpb.2023.107816](https://doi.org/10.1016/j.cmpb.2023.107816).
- [WLA*19] WANG G., LI W., AERTSEN M., DEPREST J., OURSELIN S., VERCAUTEREN T.: Aleatoric uncertainty estimation with test-time augmentation for medical image segmentation with convolutional neural networks. Neurocomputing 338 (2019), 34–45. doi:[10.1016/j.neucom.2019.01.103](https://doi.org/10.1016/j.neucom.2019.01.103).
- [WLZ*23] WANG D.-B., LI L., ZHAO P., HENG P.-A., ZHANG M.-L.: On the pitfall of mixup for uncertainty calibration. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) (2023), pp. 7609–7618. doi:[10.1109/CVPR52729.2023.00735](https://doi.org/10.1109/CVPR52729.2023.00735).
- [WSL*21] WANG D., SHELHAMER E., LIU S., OLSHAUSEN B., DARRELL T.: Tent: Fully test-time adaptation by entropy minimization. In Proceedings of the International Conference on Learning Representations (ICLR) (2021). doi:[10.48550/arXiv.2006.10726](https://doi.org/10.48550/arXiv.2006.10726).
- [YBMJ22] YU Y., BATES S., MA Y., JORDAN M. I.: Robust calibration with multi-domain temperature scaling. In Advances in Neural Information Processing Systems (NeurIPS) (2022), vol. 35, pp. 25016–25029. doi:[10.48550/arXiv.2206.02757](https://doi.org/10.48550/arXiv.2206.02757).
- [YSW*23] YANG J., SHI R., WEI D., LIU Z., ZHAO L., KE B., PFISTER H., NIETHAMMER M.: Medmnist v2 - a large-scale lightweight benchmark for 2d and 3d biomedical image classification. Scientific Data 10, 1 (2023), 41. doi:[10.1038/s41597-022-01721-8](https://doi.org/10.1038/s41597-022-01721-8).

Thank you
Questions & Discussion